

Biología de sistemas

**Intro teórica a modelos metabólicos y
aplicaciones en ingeniería metabólica**

Ingeniería metabólica clásica

De a una modificación a la vez

Un gen

Una enzima

Una vía metabólica

Limitaciones?

Visión limitada de los efectos

Modificaciones clave pueden no ser consideradas

Nuevos avances que desbloquean posibilidades

Avance de la biología sintética

Varios genes a la vez

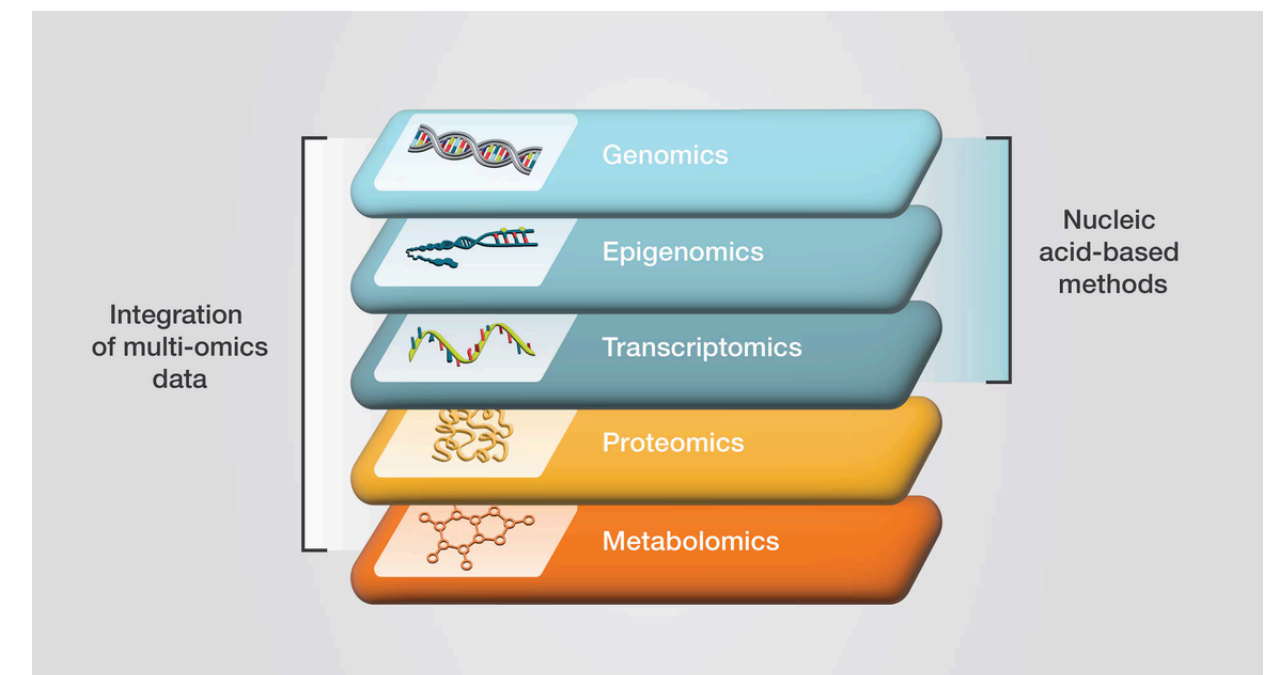
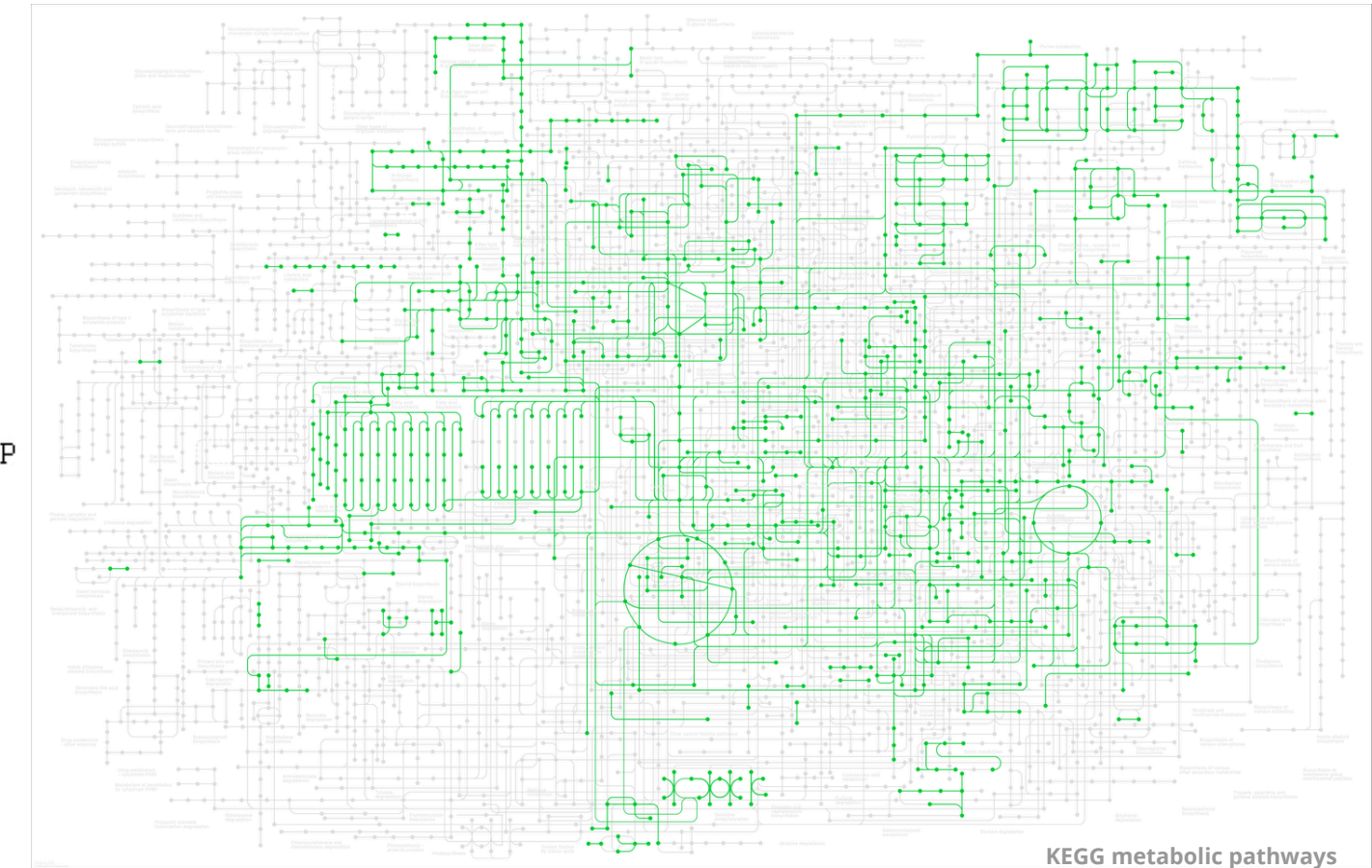
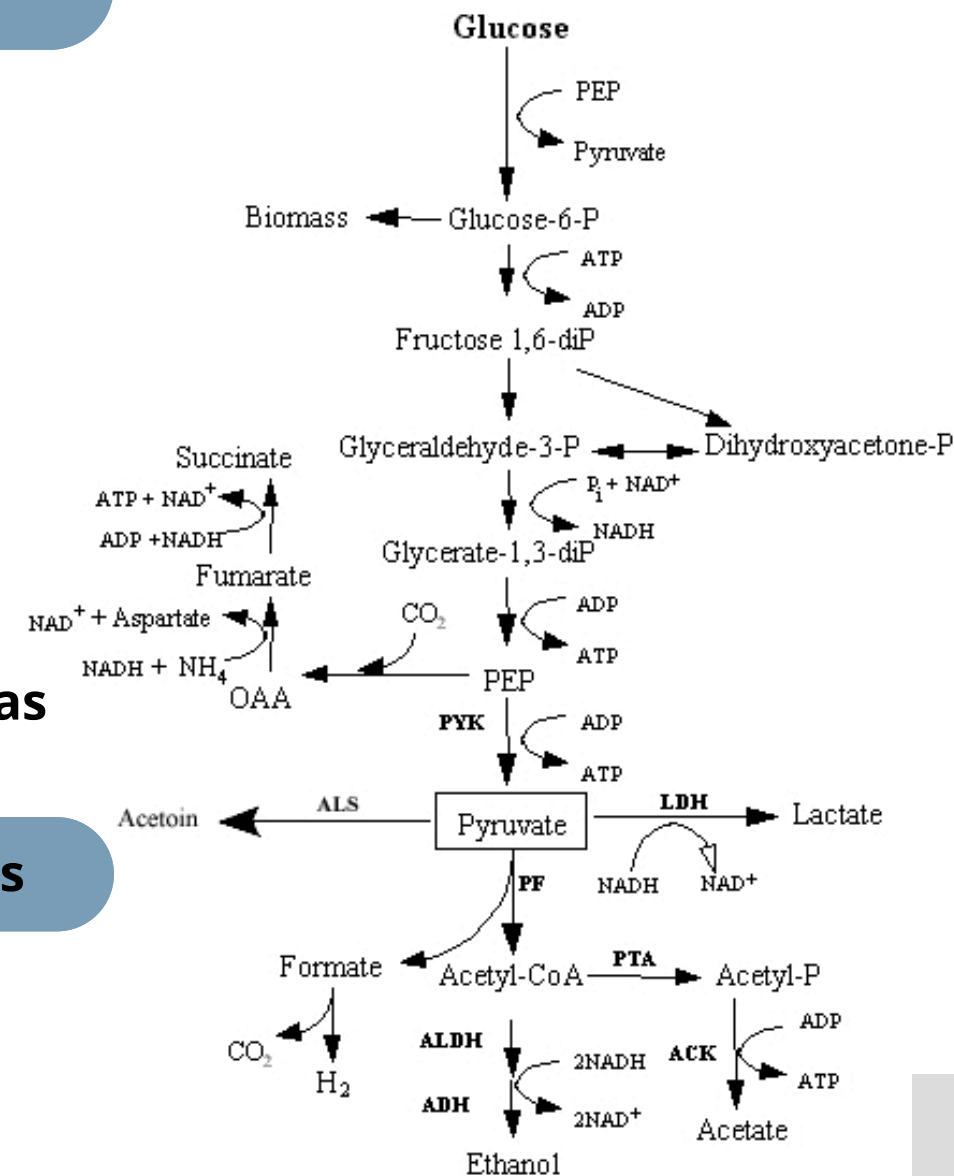
Experimentos High Throughput

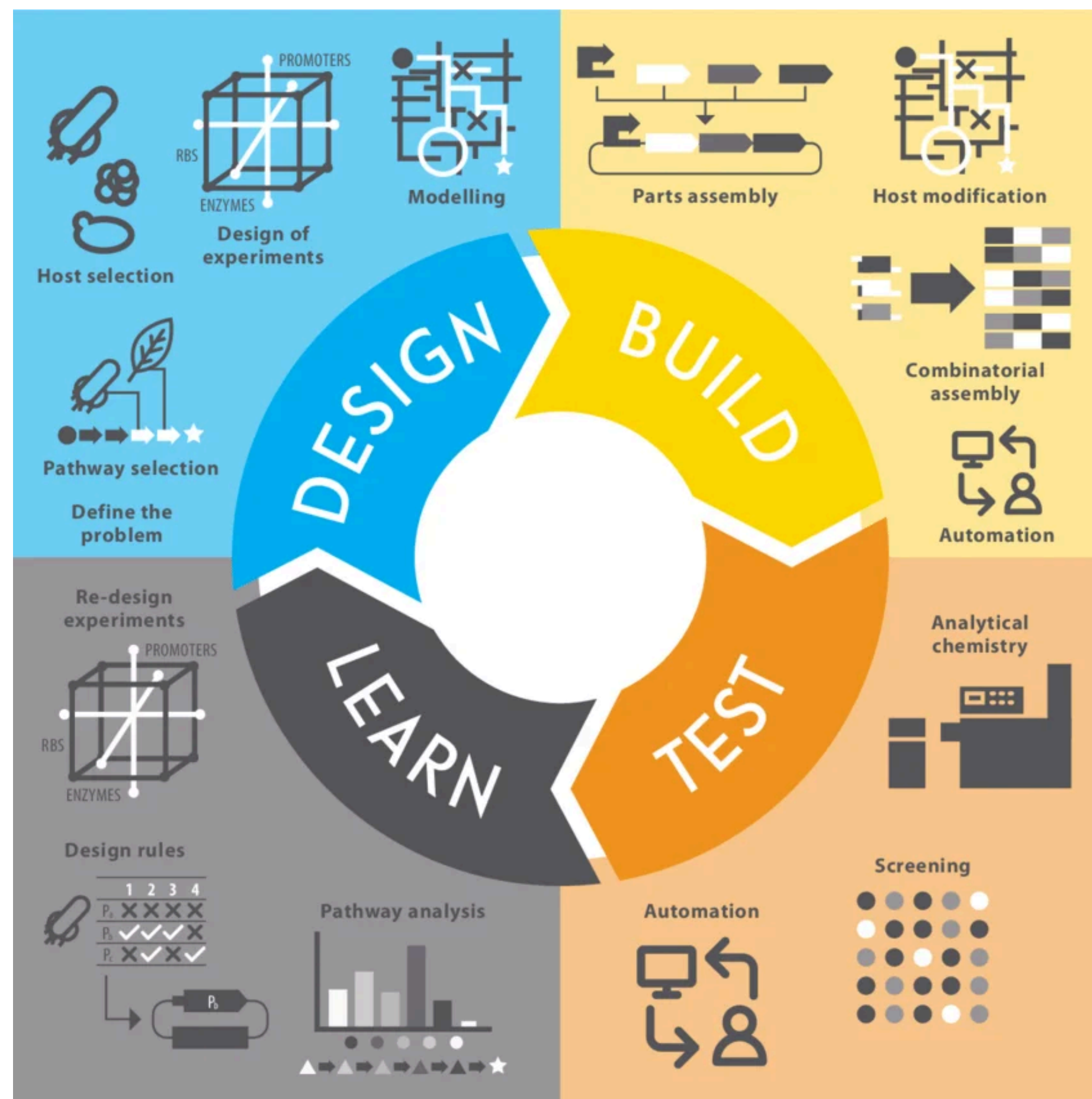
Avances computacionales

Omicas

Genome scale models

IA/ML





Genome scale metabolic models

- Mapa metabolico formalizado matematicamente para un organismo en particular
- De dónde sale?
- Dónde están?

BiGG Models

[Home](#) [Advanced Search](#) [Data Access](#) [Memote Validator](#)

Model: iML1515

Organism:

Escherichia coli str. K-12 substr. MG1655

Genome:

[NC_000913.3](#)

Model metrics:

Component	Count
Metabolites	1877
Reactions	2712
Genes	1516

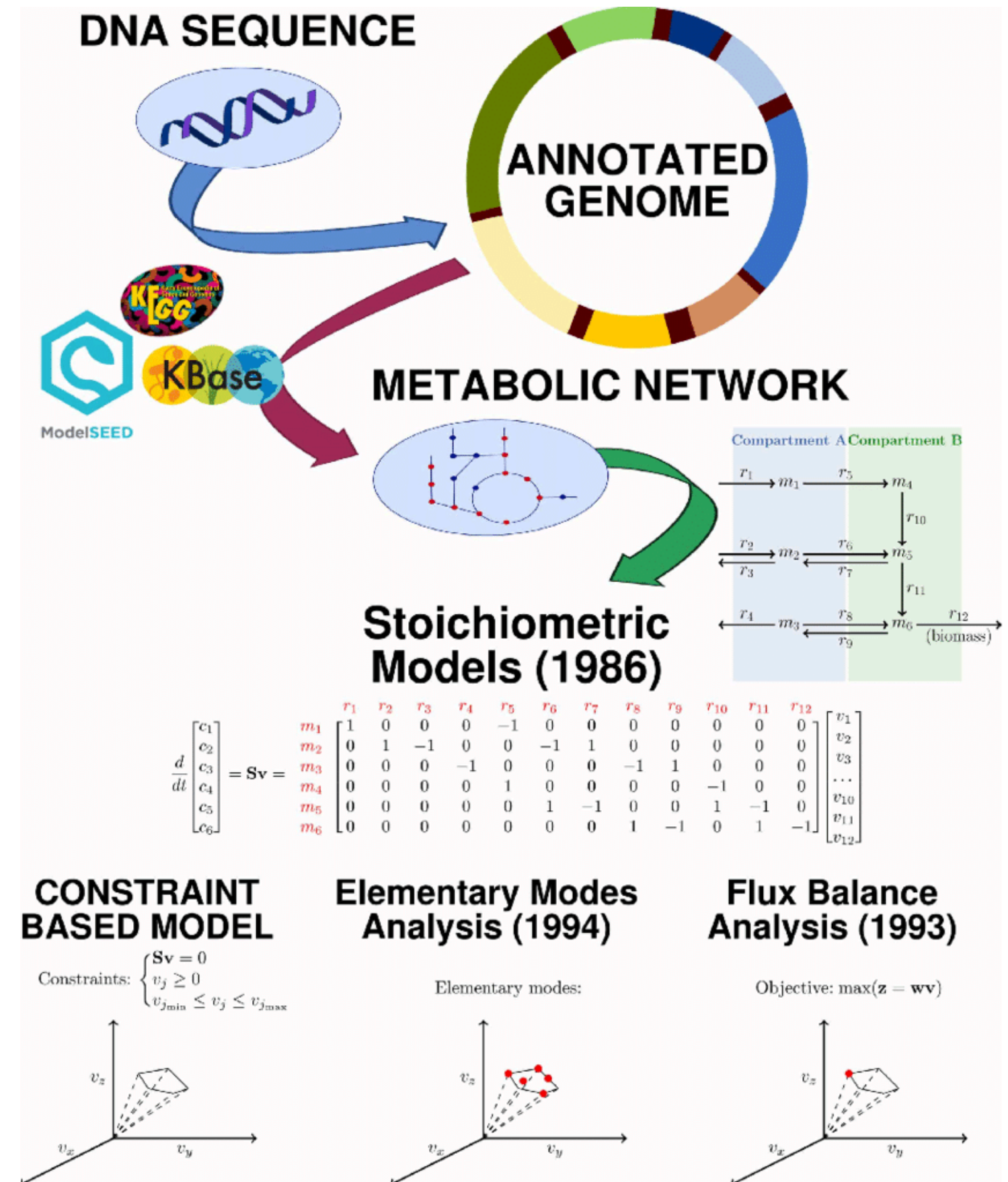
Download COBRA model from the BiGG Database:

SBML [?](#) : [iML1515.xml](#) (.xml.gz, compressed)

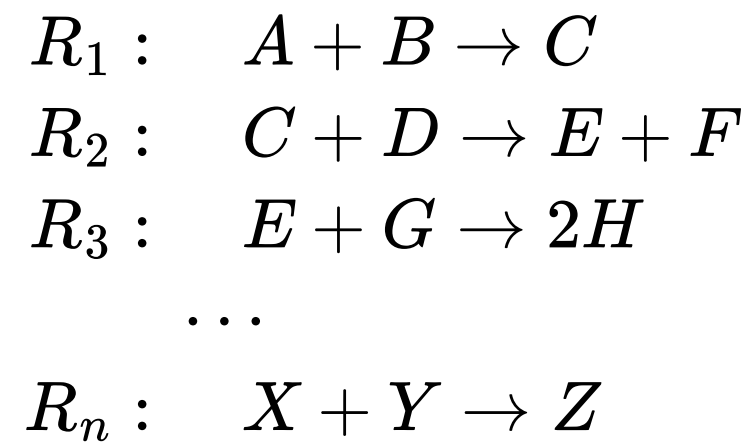
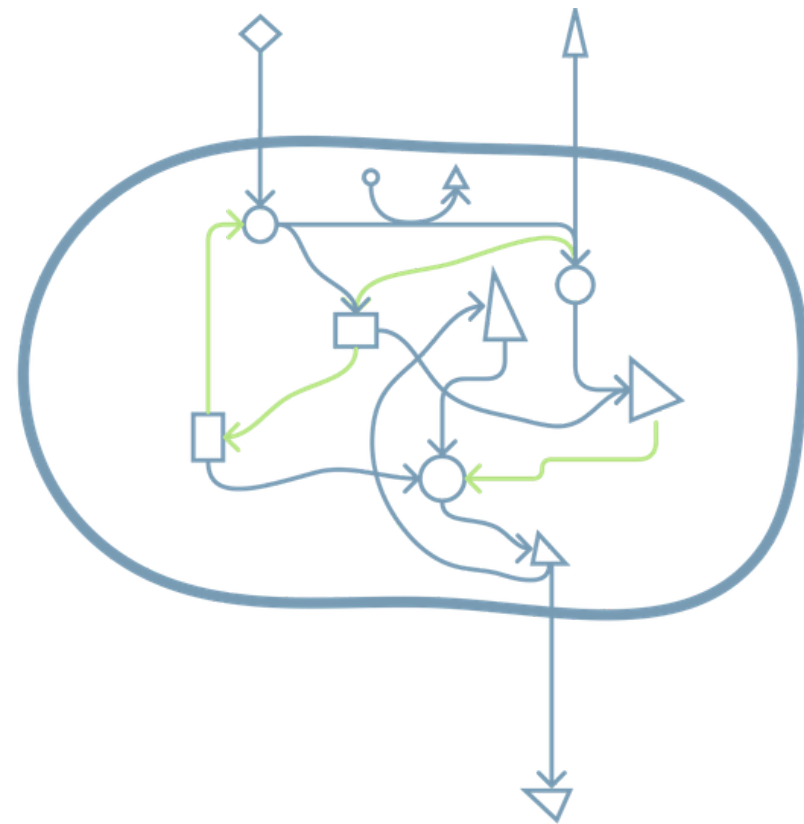
JSON [?](#) : [iML1515.json](#) (.json.gz, compressed)

MAT [?](#) : [iML1515.mat](#) (.mat.gz, compressed)

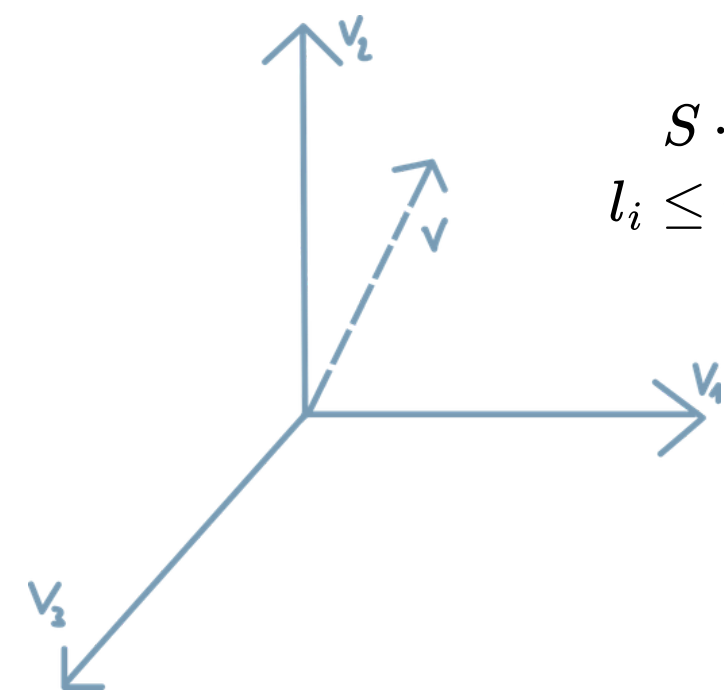
Downloads last updated Oct 31, 2019 | [BiGG License](#)



Genome-scale metabolic models

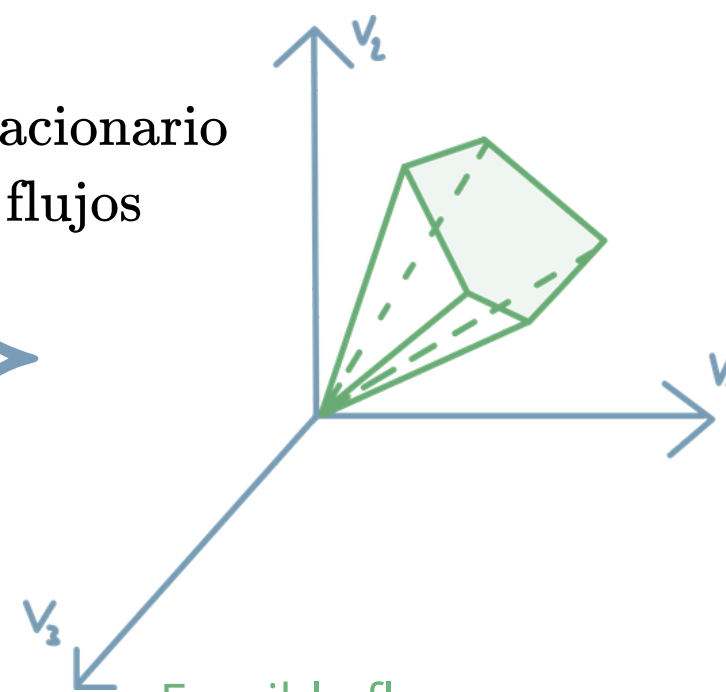


$$S = \begin{bmatrix} R_1 & R_2 & R_3 & \dots & R_n \\ A & -1 & 0 & 0 & \dots & 0 \\ B & -1 & 0 & 0 & \dots & 0 \\ C & +1 & -1 & 0 & \dots & 0 \\ D & 0 & -1 & 0 & \dots & 0 \\ E & 0 & +1 & -1 & \dots & 0 \\ F & 0 & +1 & 0 & \dots & 0 \\ G & 0 & 0 & -1 & \dots & 0 \\ H & 0 & 0 & +2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ X & 0 & 0 & 0 & \dots & -1 \\ Y & 0 & 0 & 0 & \dots & -1 \\ Z & 0 & 0 & 0 & \dots & +1 \end{bmatrix} \quad v = \begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ \vdots \\ R_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix}$$



$$\begin{aligned}
 S \cdot v &= 0 \\
 l_i &\leq v_i \leq u_i
 \end{aligned}$$

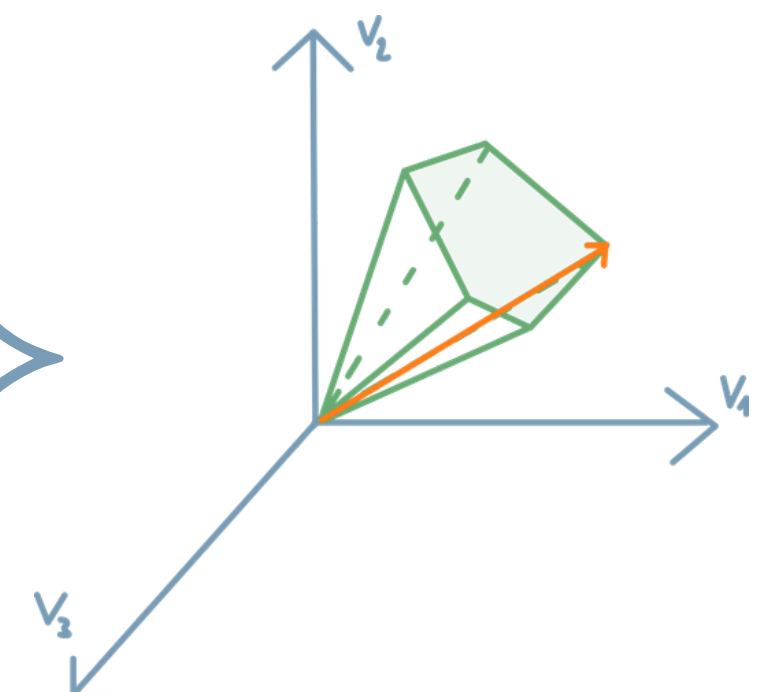
estado estacionario
limites en flujos



Feasible flux space

$$\max_v v_{\text{bio}}$$

FBA



Cómo seguimos?



Change Current Database

Current Database: [MetaCyc](#)

Search in Current Database:

Enter a gene, protein, metabolite or pathway...



BiGG Models

[Home](#) [Advanced Search](#) [Data Access](#) [Memote Validator](#)

Model: iML1515

Organism:

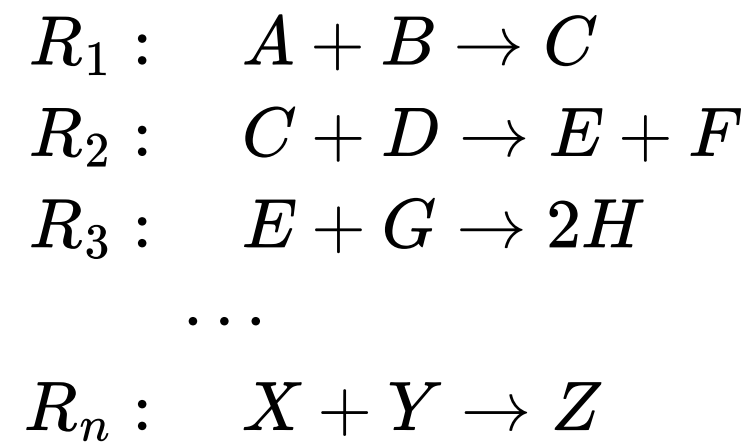
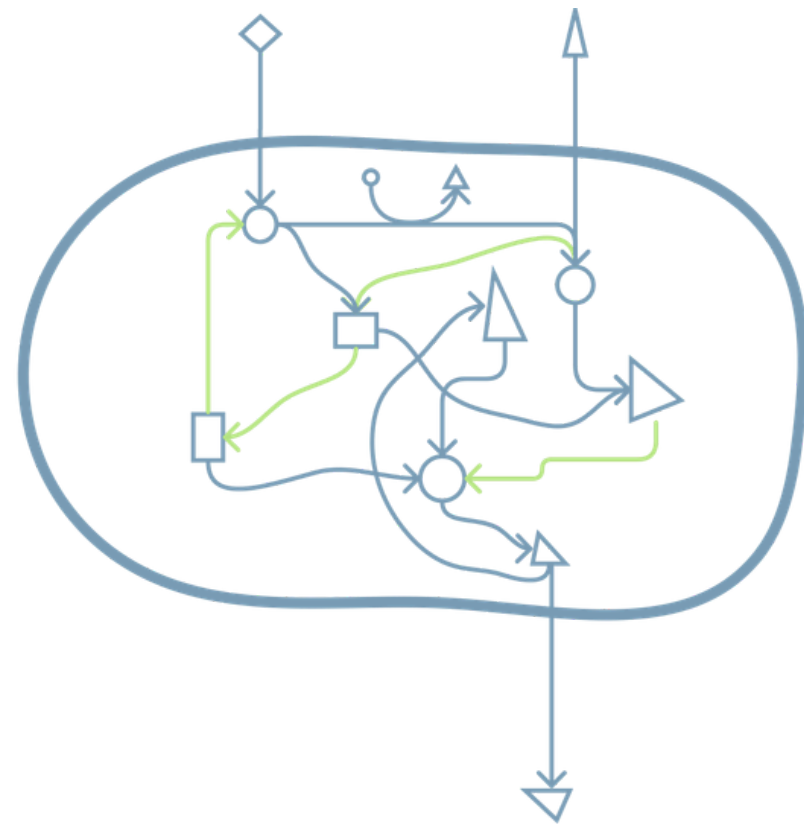
Escherichia coli str. K-12 substr. MG1655

Genome:

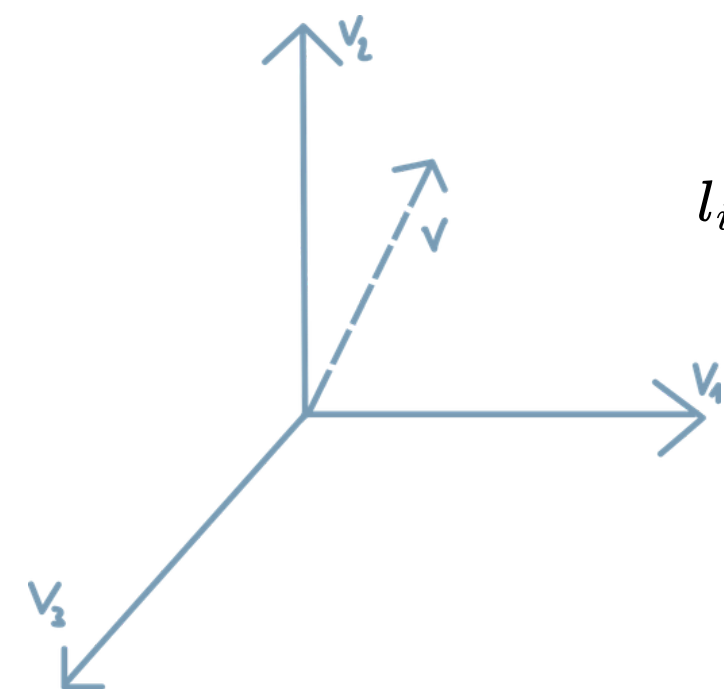
[NC_000913.3](#)



Genome-scale metabolic models

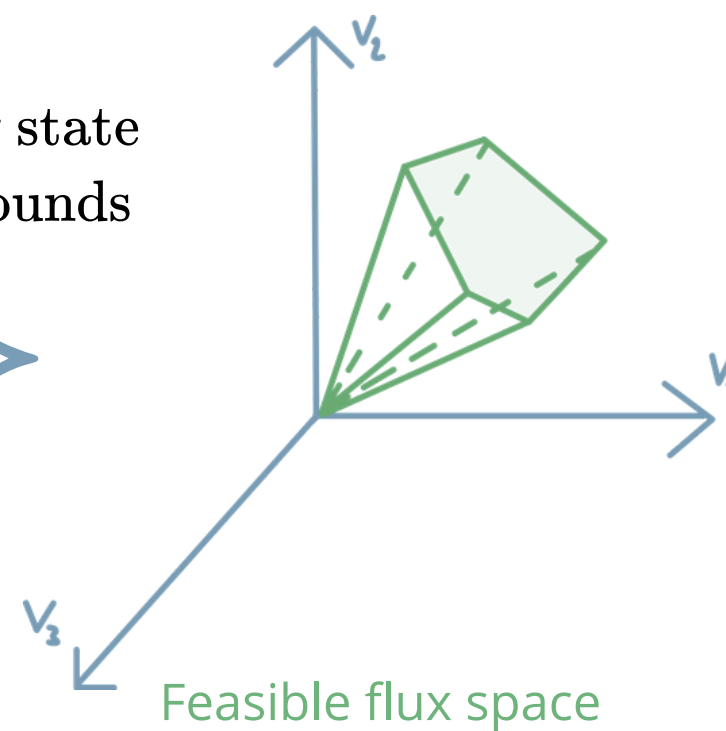


$$S = \begin{bmatrix} R_1 & R_2 & R_3 & \dots & R_n \\ A & -1 & 0 & 0 & \dots & 0 \\ B & -1 & 0 & 0 & \dots & 0 \\ C & +1 & -1 & 0 & \dots & 0 \\ D & 0 & -1 & 0 & \dots & 0 \\ E & 0 & +1 & -1 & \dots & 0 \\ F & 0 & +1 & 0 & \dots & 0 \\ G & 0 & 0 & -1 & \dots & 0 \\ H & 0 & 0 & +2 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ X & 0 & 0 & 0 & \dots & -1 \\ Y & 0 & 0 & 0 & \dots & -1 \\ Z & 0 & 0 & 0 & \dots & +1 \end{bmatrix} \quad v = \begin{bmatrix} R_1 \\ R_2 \\ R_3 \\ \vdots \\ R_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_n \end{bmatrix}$$



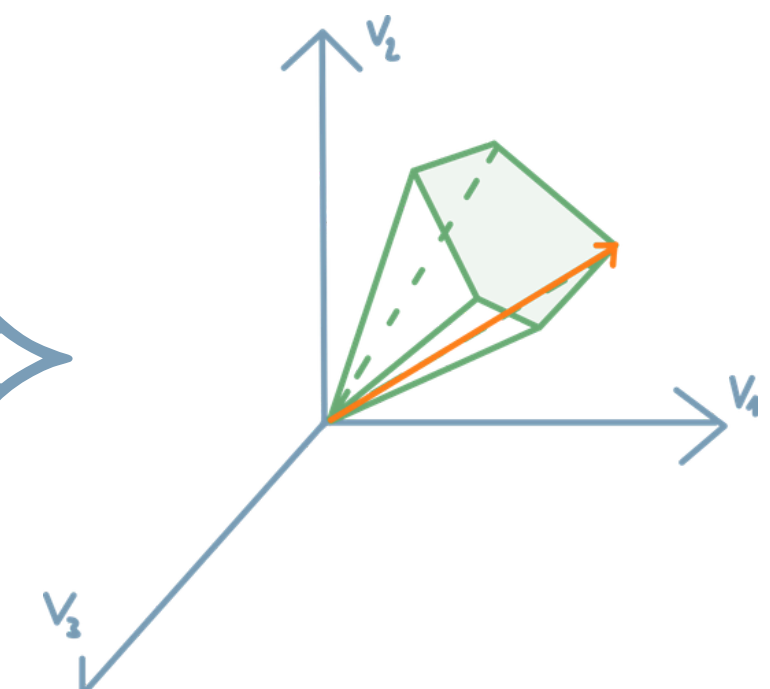
$$\begin{aligned}
 S \cdot v &= 0 \\
 l_i &\leq v_i \leq u_i
 \end{aligned}$$

steady state
flux bounds



$$\max_v v_{\text{bio}}$$

FBA



Cómo podemos usar esto para ingeniería metabólica?

Computational strain design reshapes the feasible flux space



Computational strain design searches for genetic interventions that change which phenotypes are possible

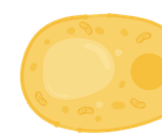
→ **Gene KI** → New reaction(s) → **Add** new functions to network

→ **Gene KO** → Remove reaction(s) → **Remove** native functions from network

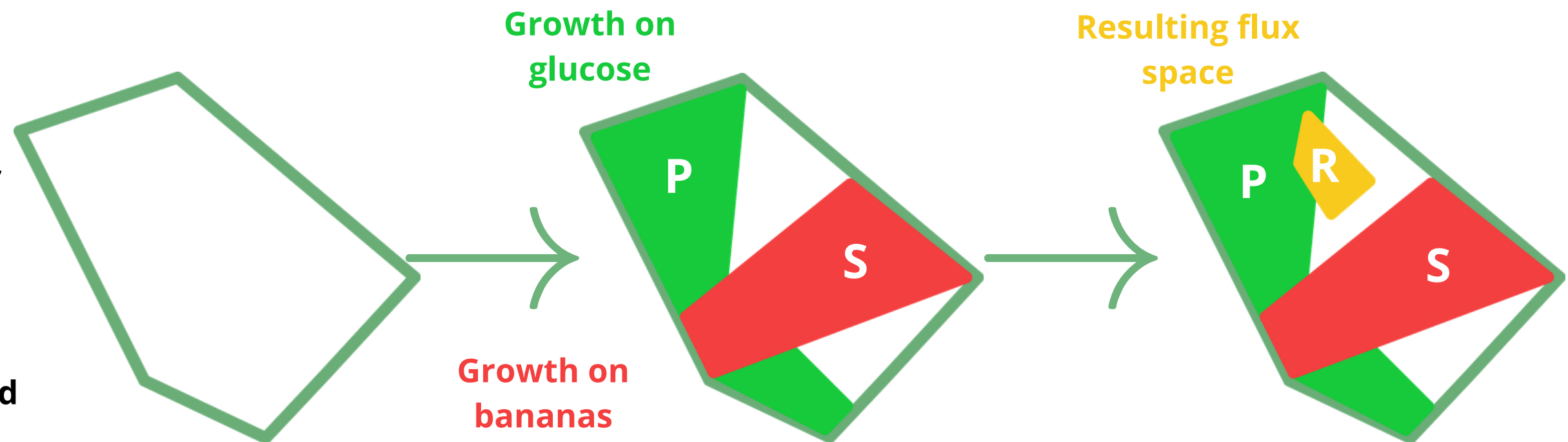
StrainDesign: a systematic way to find these interventions

Existing method by Schneider et al. 2022

 Generalized Minimal Cut Sets algorithm

 Encode desired phenotype as regions of the flux space to **PROTECT** or **SUPPRESS**

 Find **irreducible** intervention sets that keep the **desired** phenotype feasible and make **undesired** phenotypes infeasible



Flexible but underused framework.

Improved xylose and arabinose utilization by an industrial recombinant *Saccharomyces cerevisiae* strain using evolutionary engineering

Research | [Open access](#) | Published: 15 June 2010
Volume 3, article number 13 (2010) [Cite this article](#)

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Rosa Garcia Sanchez, Kaisa Karhumaa, César Fonseca, Violeta Sánchez Nogué, João RM Almeida, Christer U Larsson, Oskar Bengtsson, Maurizio Bettiga, Bärbel Hahn-Hägerdal & Marie F Gorwa-Grauslund 

Laboratory evolution for forced glucose-xylose co-consumption enables identification of mutations that improve mixed-sugar fermentation by xylose-fermenting *Saccharomyces cerevisiae* 

[Ioannis Papapetridis](#), [Maarten D Verhoeven](#), [Sanne J Wiersma](#), [Maaïke Goudriaan](#), [Antonius J A van Maris](#), [Jack T Pronk](#)  [Author Notes](#)

Engineering of *Saccharomyces cerevisiae* for Efficient Anaerobic Alcoholic Fermentation of L-Arabinose

Authors: [H. Wouter Wisselink](#), [Maurice J. Toirkens](#), [M. del Rosario Franco Berriel](#), [Aaron A. Winkler](#), [Johannes P. van Dijken](#), [Jack T. Pronk](#), [Antonius J. A. van Maris](#)  | [AUTHORS INFO & AFFILIATIONS](#)

Enhancing the Co-utilization of Biomass-Derived Mixed Sugars by Yeasts



Meirong Gao ^{1,2}



Deon Ploessl ^{1,2}



Zengyi Shao ^{1,2,3,4*}

Engineering of Pentose Transport in *Saccharomyces cerevisiae* for Biotechnological Applications

Jeroen G. Nijland and Arnold J. M. Driessen *

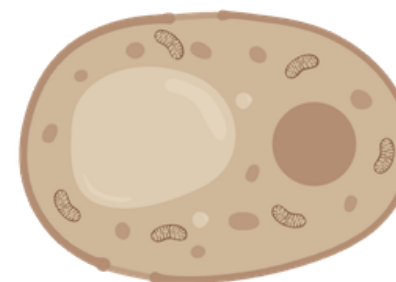
Current strategies and challenges

→ *S. cerevisiae* consumes **glucose** preferentially, and pentose sugars slowly, inefficiently or not at all

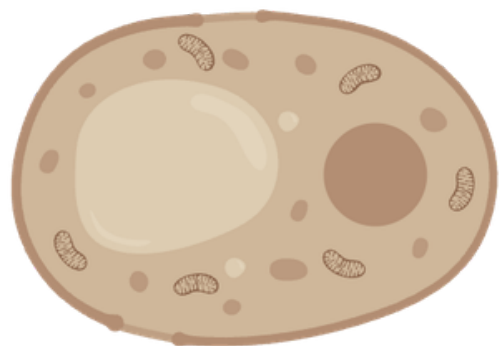
→ Engineering strategies:

- Introducing heterologous pathways for pentose assimilation
- Transporter engineering
- Adaptive laboratory evolution

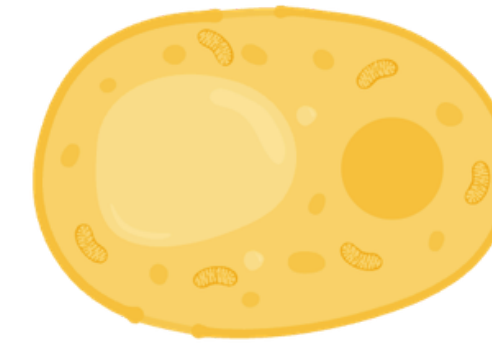
→ Enabling co-consumption is not enforcing co-consumption



Instead of only enabling co-consumption, can we make co-consumption required for growth?



Find KO/KI sets that make **growth possible only when all three sugars are consumed**.
No feasible flux distribution should remain where growth is supported by only one sugar or by any pair of sugars.



Conditions

- **Model: Yeast9**
- **Aerobic or Anaerobic setup**
- **Glucose, Xylose, Arabinose uptake allowed**

KI and KO candidates

- **KI candidates:**
13 heterologous reactions for xylose and arabinose metabolism.
- **KO candidates:**
all native genes in Yeast9 model

SUPPRESS modules (x9)

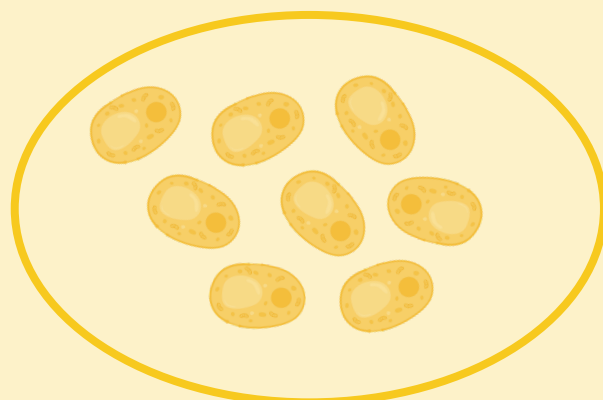
- **SUPPRESS** all flux distributions where:
 - Growth + uptake of one sugar
 - Growth + uptake of two sugars
 - Growth + marginal uptake of sugar

PROTECT module (x1)

- **PROTECT** at least one flux distribution where:
 - Growth + uptake of all three sugars

Simulations

- 30 independent StrainDesign runs
- 30 min time limit
- 32 CPU cores
128 GB of memory
- Max intervention cost: 55



Verify, rank and analyze designs

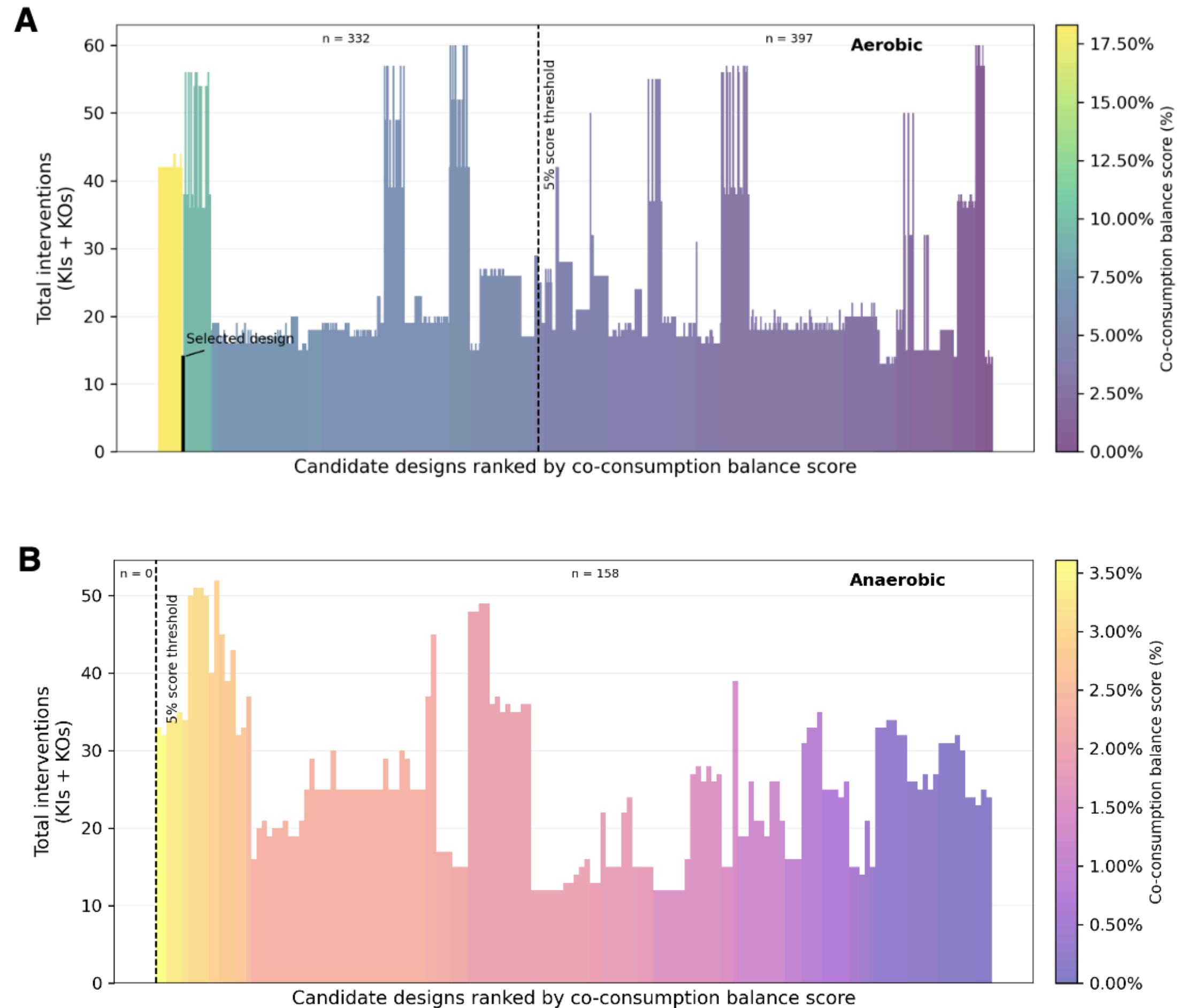
- Balance score = smallest sugar contribution across all sugars and uptake-minimization orders.
- Decomposition into eCAT, pCAT and ANA following Remeijer et al.

Overall results

- 729 aerobic designs
- 158 anaerobic designs
- 13 interventions are enough to enforce co-consumption
- 332 aerobic strains have a balance score > 5%.
- No anaerobic strain reaches a 5% balance score. Industrially relevant challenge.

StrainDesign finds intervention sets that enforce co-consumption.

Computationally impossible to enumerate all solutions. The simulations ran give a sample of the solution space.



One selected aerobic design

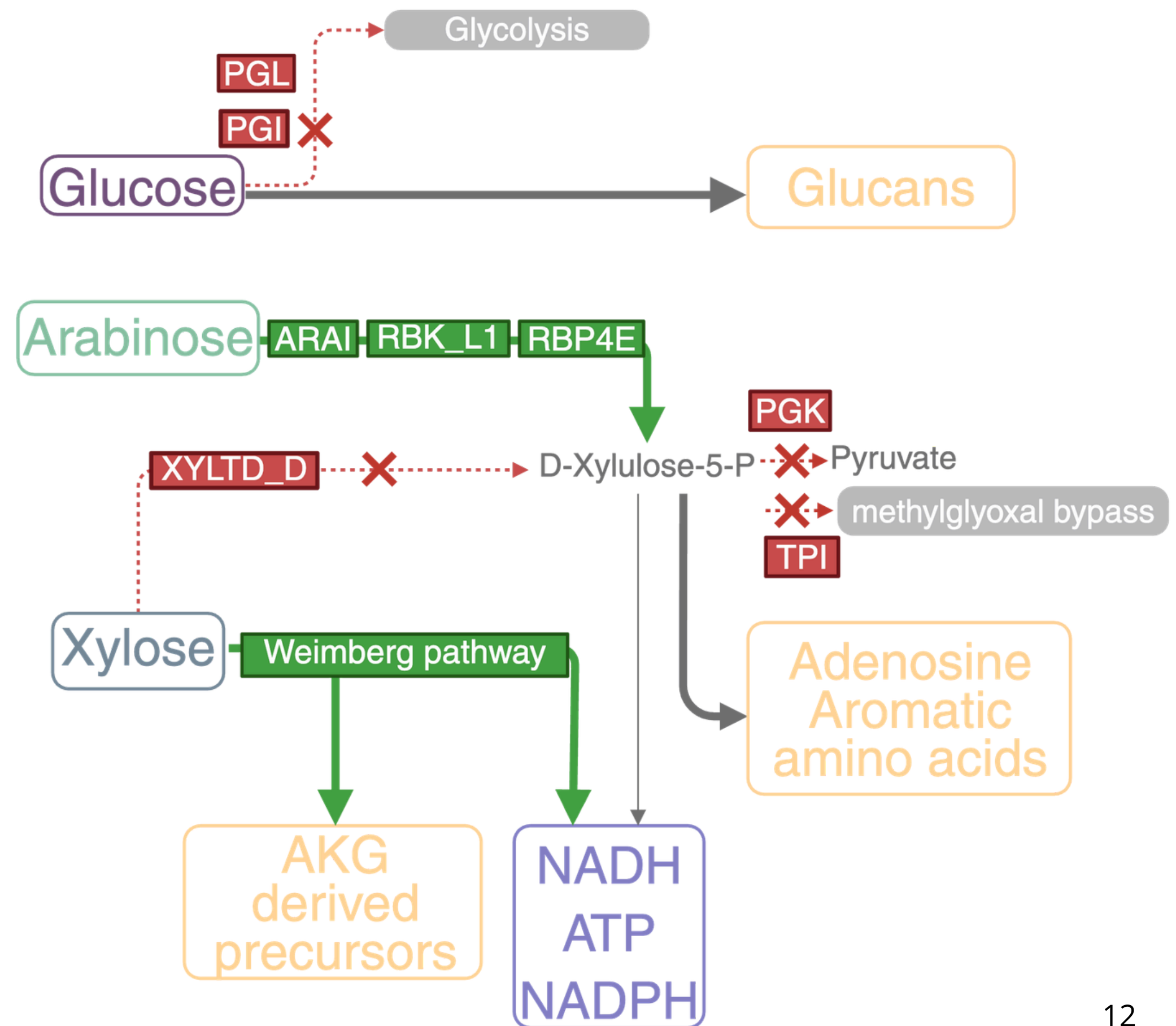
→ KIs introduce pentose-assimilation routes:

- **Arabinose** → Bacterial isomerase pathway
- **Xylose** → Weimberg pathway

→ KOs block alternative routes:

- **PGL, PGI** block entrance of **glucose** to glycolysis
- **PGK, TPI** block **arabinose**-derived pyruvate production and ATP production through Methylglyoxal bypass.
- **XYLTD_D** blocks entry of **xylose** to PPP.
- **FDH** blocks alternative pathways that allow growth without **xylose**

Precursors and **charged energy carriers** made by one sugar cannot be fully replaced by the others. The three sugars become complementary.



Landscape of solutions

→ Some candidate KIs were never selected in sampled designs

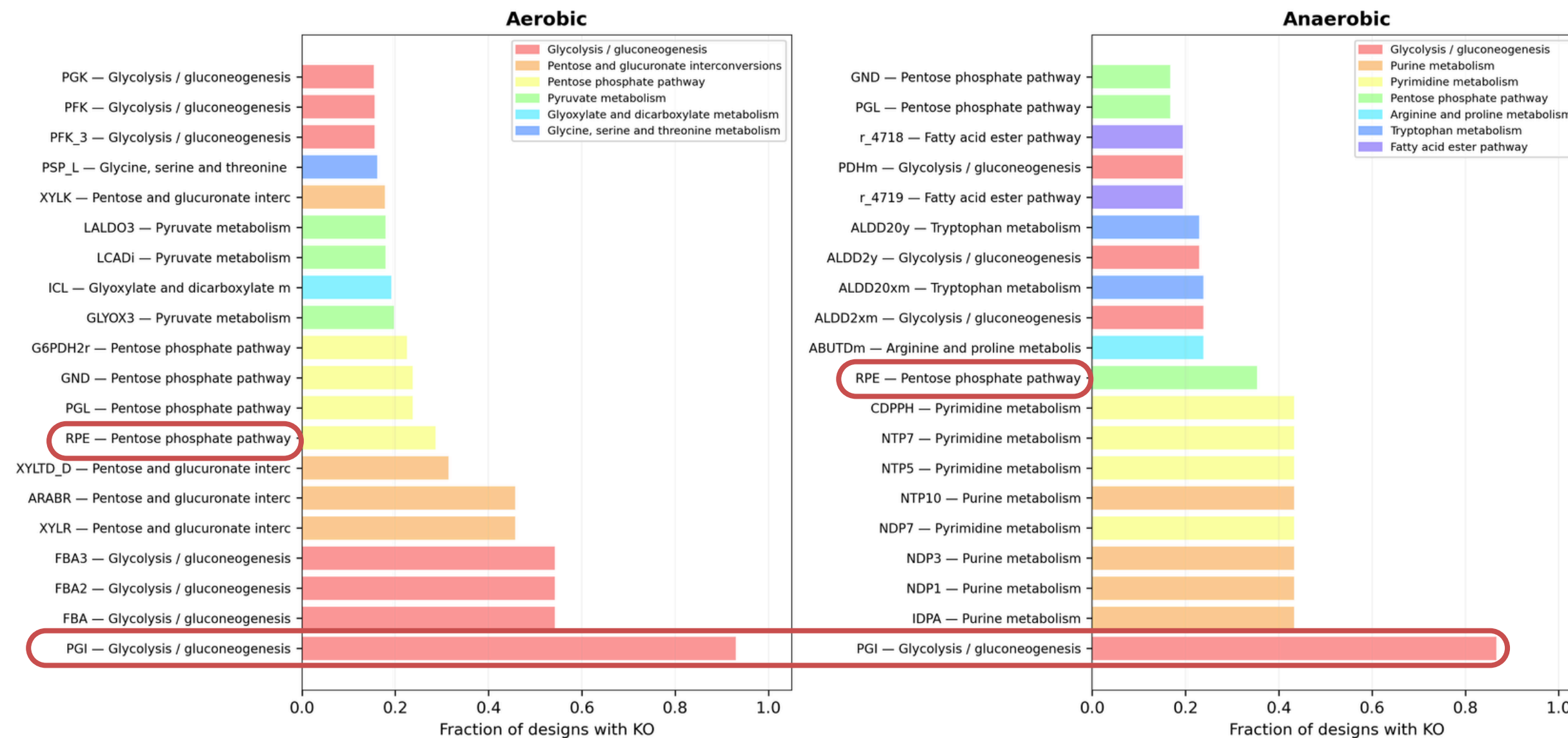
→ PGI most frequent KO in anaerobic and aerobic designs

→ RPE also appears as frequent KO

→ Not all KOs have an obvious effect from pathway inspection

Design of an *S. cerevisiae* strain whose growth depended on extensive co-consumption of xylose and glucose was based on the observation that inactivation of *PGI1* blocks entry of glucose-6-phosphate into glycolysis, while inactivation of *RPE1* prevents entry of ribulose-5-phosphate into the non-oxidative PPP (Fig. 1). As a consequence, a *pgi1Δ rpe1Δ* strain is unable to grow on glucose. If conversion of xylose into xylulose-5-

Papapetridis et al. 2018



Known KO logic appears in the solution set, but this method also reveals non-obvious targets.

By “human” analysis of pathway maps, these would be most likely impossible to find.

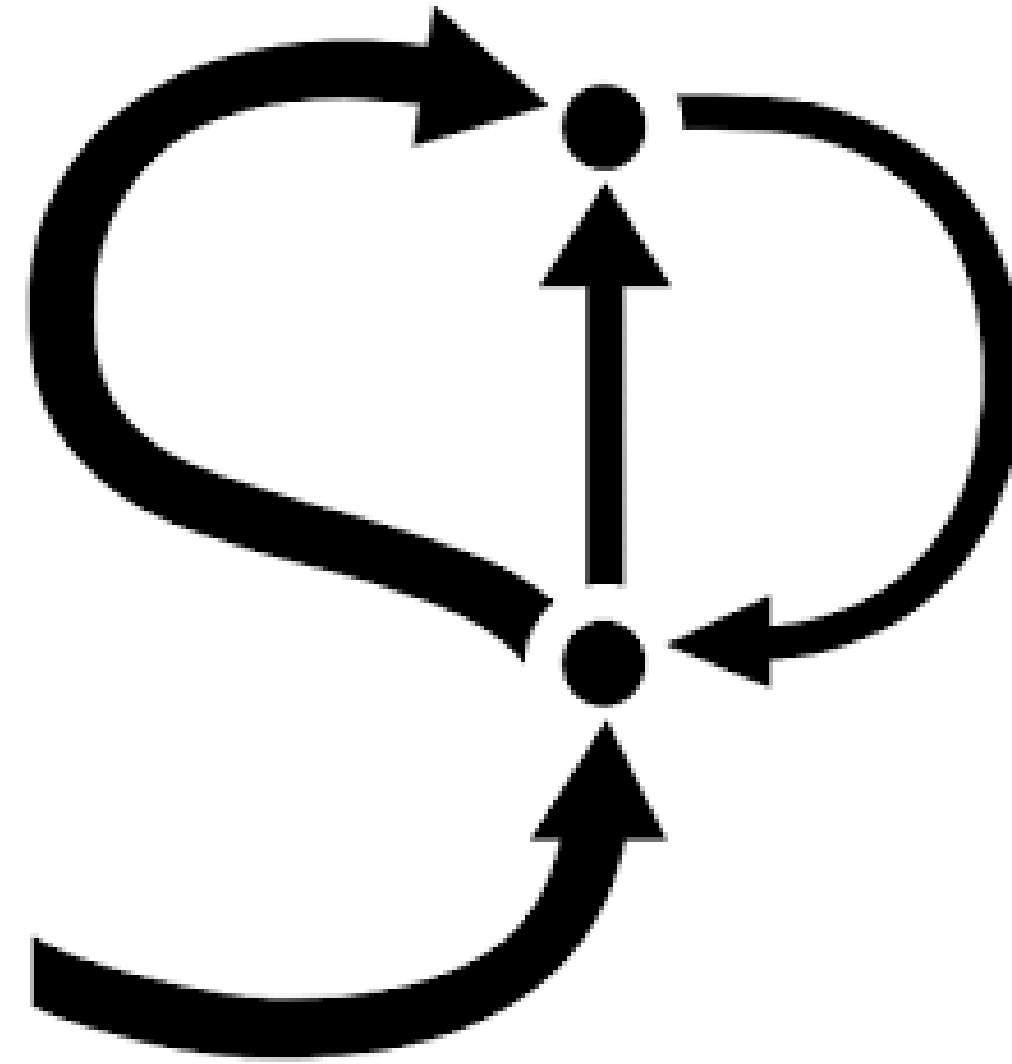
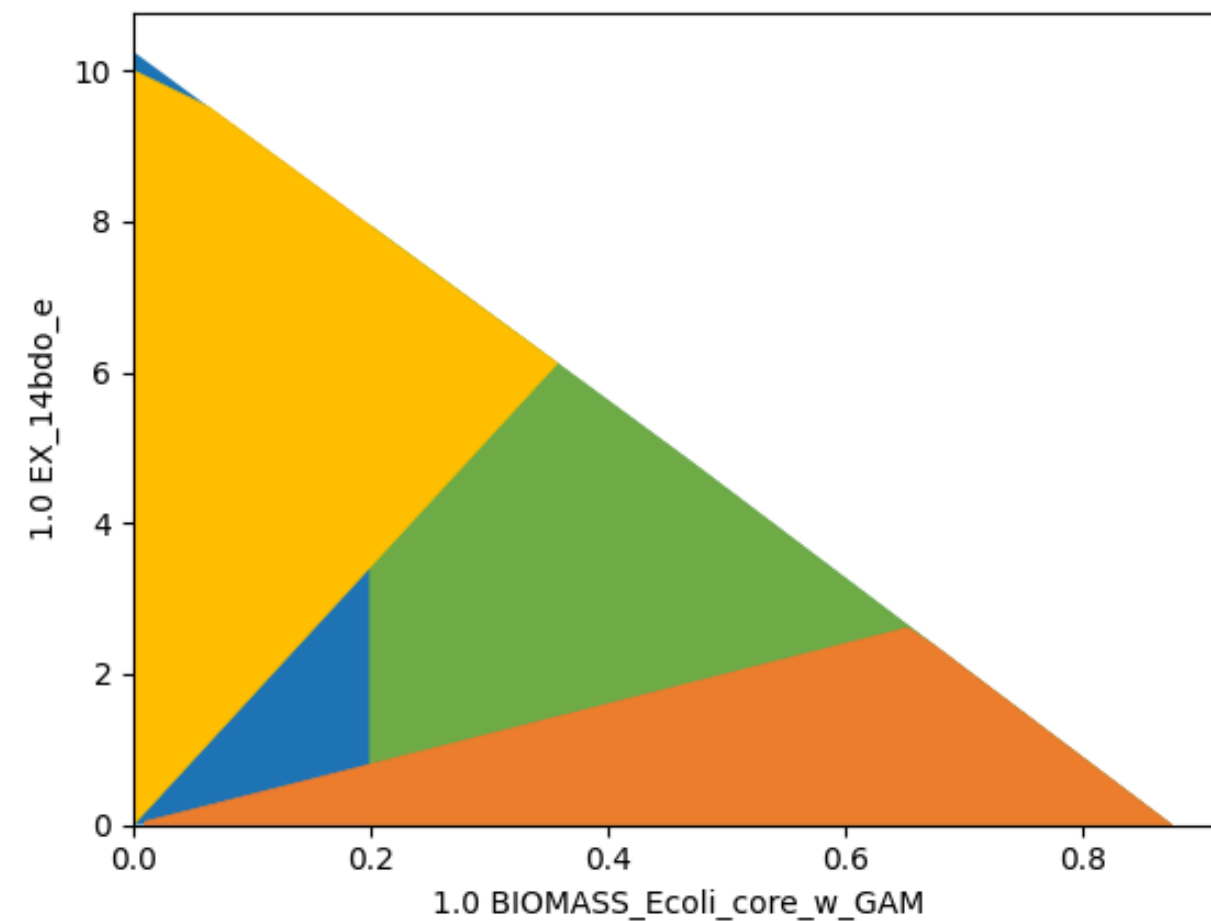
Otras aplicaciones de StrainDesign

Cualquiera que se nos ocurra!

Pasos:

- Lista de reacciones
- Agregarlas al modelo
- Definir los fenotipos deseados
- Correr el modelo

Por ejemplo: fijación de CO₂ en *E. coli*



These type of methods do not reach the experimental metabolic engineers. There is a gap to bridge.

 PATHWAY ENGINEERING & CO-CONSUMPTION ASSISTANT

1 · Host2 · Sugars3 · Candidates4 · Settings5 · Run

CURRENT CONFIG

Model
—

Sugars
none

KI candidates
0 reactions

KO candidates
All gene rules

SOLVER

Backend
glpk

Max cost / solutions
10 / 10

STATUS

Idle

Choose host model

Select a preloaded genome-scale model or upload your own SBML/JSON file. The model defines all available reactions and genes.

● Preloaded models

☐ PECA_toy.xml

☐ iMM904.xml

☐ yeast9_anaerobic_biggids.xml

● Upload your own model

↑

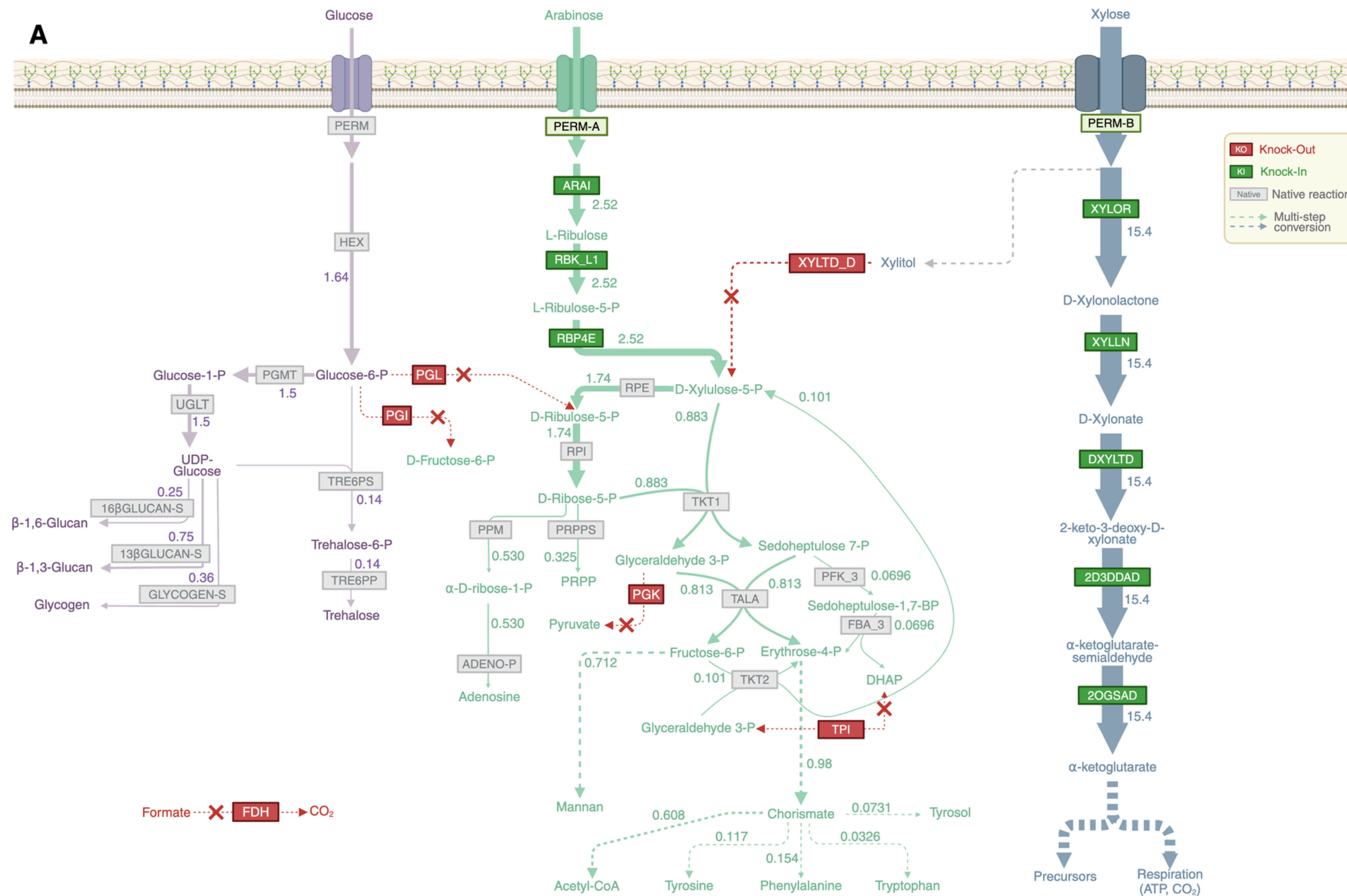
Drop SBML (.xml) or COBRA JSON (.json) here, or click to browse

Next: Choose sugars →



Limitaciones

Oportunidades



B